Total Questions: 35

LEVEL – II

Time: 1.5 hr.
Total Marks: 70

SECTION - 1	• This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer. • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.
SECTION - 2	• This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s). • For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given. • There are **NO NEGATIVE** marks for wrong answers.
SECTION - 3	• This section contains **NINE (09)** questions of PASSAGE TYPE OR COMPREHENSION TYPE. • Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer. • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for
SECTION - 4	• This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS.** • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. **A shopkeeper sells pencils at ₹3 each and erasers at ₹2 each. If a student buys x pencils and y erasers, which expression shows total cost?**

(A) $3x + 2y$ (B) $5xy$

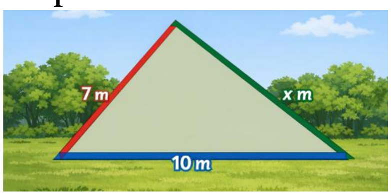
(C) $3x + y$ (D) $2x + 3y$

2. **A shop sells 12 pens for Rs. 180. During a sale, a student buys 18 pens and gets a 10% discount on total cost. How much does he pay?**

(A) Rs. 243 (B) Rs. 250

(C) Rs. 270 (D) Rs. 300

3. **A triangular park has sides of lengths 7 m, 10 m, and x m. The park must form a valid triangle. Which of the following values of x is possible?**



(A) 3 (B) 17

(C) 5 (D) 20

4. A triangle-shaped garden has base 12 m and height 8 m. What is the cost of grassing at ₹5 per m^2?

(A) ₹240 (B) ₹480

(C) ₹360 (D) ₹300

5. A stretched wire between two poles has fixed ends and cannot extend further. Which geometric concept best represents it?

(A) Line (B) Ray

(C) Line segment (D) Point

6. The sum of a father's age and his son's age is 50 years. The father is 4 times as old as the son. Find the son's age.

(A) 10 (B) 12

(C) 8 (D) 15

7. A shopkeeper marks a discount as 2/5 of the price. Which of the following represents the same discount?

(A) 4/10 (B) 6/20

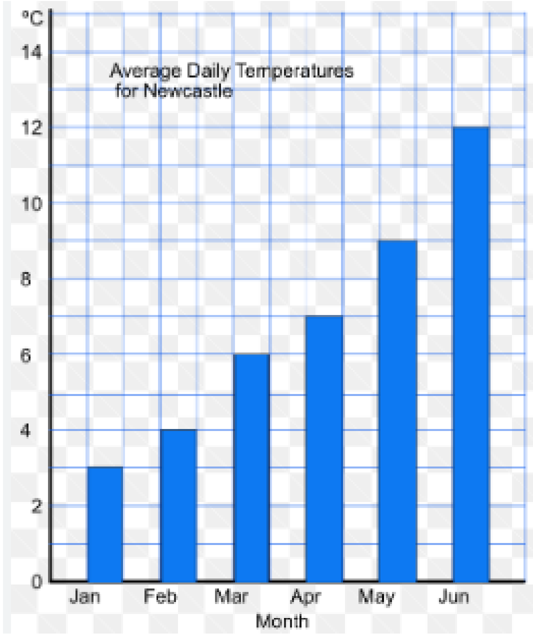
(C) 8/25 (D) 10/30

8. A water tank contains 50 L of water. After using 12.735 L, how much water remains?

(A) 37.265 L (B) 38.265 L

(C) 37.735 L (D) 36.265 L

9. Read the bar graph and answer the following question.



What is the average temperature from January to March?

(A) 3°C (B) 4°C

(C) 4.33°C (D) 5°C

10. A person walks 8 steps to the right from 0 and then 13 steps to the left. Where is he now?

(A) –5 (B) 5

(C) –21 (D) 21

11. Two sides of a triangle are 9 cm and 14 cm. The third side must be:

(A) less than 5 cm

(B) greater than 23 cm

(C) between 5 cm and 23 cm

(D) exactly 5 cm

12. A machine works in two stages: first multiplies output by 2^3, then by 2^2. Total effect is:

(A) 2^5 (B) 2^6

(C) 4^5 (D) 2^1

13. A farmer uses $\frac{3}{5}$of his land for crops. Out of that, $\frac{2}{3}$is used for wheat.

What fraction of total land is used for wheat?

(A) $\frac{7}{15}$ (B) $\frac{2}{5}$

(C) $\frac{3}{8}$ (D) $\frac{1}{5}$

14. A water tank used in a house has 6 rectangular faces. Which solid shape does it represent?

(A) Cube (B) Cylinder

(C) Cuboid (D) Cone

15. The cost of 5 kg rice is Rs. 250. If the price increases by 20%, what is the cost of 8 kg rice?

(A) Rs. 480 (B) Rs. 400

(C) Rs. 360 (D) Rs. 300

Section - 2

16. In designing a corner shelf, two wooden pieces form complementary angles. Which pairs are possible?

(A) 25° and 65° (B) 40° and 50°

(C) 80° and 20° (D) 100° and −10°

17. The cost of 10 items is Rs. 250. Which of the following are correct?

(A) Cost of 1 item = Rs. 25

(B) Cost of 20 items = Rs. 500

(C) Cost of 5 items = Rs. 100

(D) Cost of 15 items = Rs. 375

18. A rectangle has length 18 m and breadth 6 m. Which are correct?

(A) Area = 108 m²

(B) Perimeter = 48 m

(C) Diagonal = √(360) m

(D) Area = 24 m²

19. Which expressions represent a situation correctly?

(A) ₹10 per item for $x$items → $10x$

(B) ₹5 discount on total $3x$→ $3x - 5$

(C) Double of y plus 3 → $2y + 4$

(D) Sum of x and x^2→ $2x^2$

20. A teacher records marks of students: 12, 15, 18, 10, 20. Which of the following statements are correct?

(A) Mean = 15

(B) Mean lies between smallest and largest value

(C) Mean = Median

(D) Mean changes if one value increases

21. A person has −₹200. Which transactions will make the balance zero?

(A) Deposit ₹200

(B) Withdraw ₹200

(C) Deposit ₹150 and ₹50

(D) Deposit ₹100 twice

22. Which of the following sets contain like fractions?

(A) $\frac{2}{7}, \frac{5}{7}, \frac{9}{7}$ (B) $\frac{3}{5}, \frac{6}{10}, \frac{9}{15}$

(C) $\frac{4}{9}, \frac{7}{9}, \frac{1}{9}$ (D) $\frac{8}{11}, \frac{3}{13}, \frac{5}{17}$

23. Which are correct?

(A) If $2^x = 2^5$, then $x = 5$

(B) If $3^x = 27$, then $x = 3$

(C) If $a^2 = 9$, then $a = 3$ only

(D) If $4^x = 16$, then $x = 2$

Section - 3

Passage 1:

A factory produces pens. The number of pens produced is directly proportional to the number of workers.

10 workers produce 500 pens in a day.

24. How many pens will 25 workers produce?

(A) 1000 (B) 1100

(C) 1250 (D) 1500

25. Which of the following is a correct proportion?

(A) 2:3 = 4:5 (B) 3:6 = 4:8

(C) 5:7 = 10:12 (D) 6:9 = 8:10

26. If production doubles when workers double, what type of variation is this?

(A) Inverse proportion

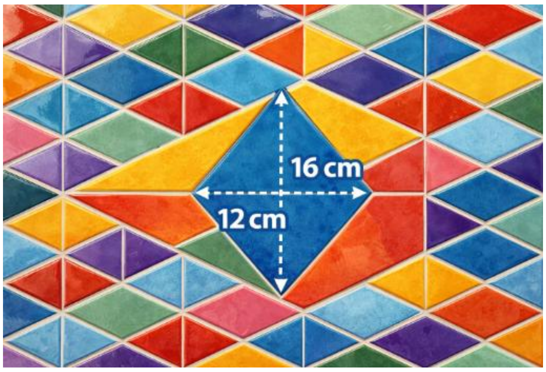
(B) Direct proportion

(C) No relation

(D) Constant ratio

PASSAGE-2

A decorative floor is made using rhombus-shaped tiles. Each tile has diagonals 12 cm and 16 cm.



27. Area of one tile is:

(A) 96 cm^2 (B) 192 cm^2

(C) 384 cm^2 (D) 48 cm^2

28. If 200 such tiles are used, total area covered is:

(A) 19200 cm^2 (B) 24000 cm^2

(C) 38400 cm^2 (D) 40000 cm^2

29. If cost per cm^2 is ₹0.5, total cost is:

(A) ₹9600 (B) ₹19200

(C) ₹20000 (D) ₹15000

PASSAGE-3

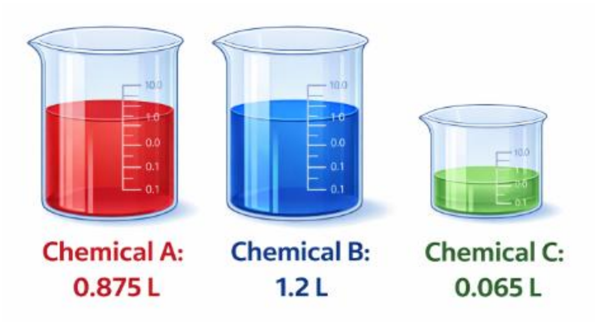
A lab prepares a solution by mixing three chemicals:

Chemical A: **0.875 L**

Chemical B: **1.2 L**

Chemical C: **0.065 L**

During evaporation, **0.145 L** is lost. The remaining solution is filled into bottles of **0.25 L each.**



30. Total initial solution volume:

(A) 2.14 (B) 2.135

(C) 2.145 (D) 2.125

31. Remaining solution after evaporation:

(A) 1.995 (B) 2.005

(C) 1.985 (D) 2.015

32. Number of bottles filled:

(A) 5 (B) 7.98

(C) 9.5 (D) 10

Section - 4

33.

Column I (Situation)	Column II (Value)
(a) 6 workers finish work in 8 days. How many days will 12 workers take?	(i) 30 hours
(b) 10 taps fill a tank in 15 hours. How many hours will 5 taps take?	(ii) 12 days
(c) 8 men finish a work in 6 days. How many days will 4 men take?	(iii) 4 days
(d) 15 workers finish work in 4 days. How many days will 3 workers take?	(iv) 20 day

(A) a–i, b–ii, c–iv, d–iii
(B) a–iii, b–i, c–iv, d–ii
(C) a–ii, b–i, c–iii, d–iv
(D) a–iii, b–i, c–ii, d–iv

34.

Column I (Situation)	Column II (Result)
(a) A square park has side 12 m. Find its perimeter	(i) 144 m^2
(b) A square tile has area 169 cm^2. Find its side	(ii) 48 m
(c) A square field has diagonal $10\sqrt{2}$ m. Find its side	(iii) 13 cm
(d) A square garden has side 12 m. Find its area	(iv) 10 m

(A) a–ii, b–i, c–iii, d–iv
(B) a–ii, b–iii, c–iv, d–i
(C) a–i, b–iv, c–iii, d–ii
(D) a–ii, b–i, c–iv, d–iii

35.

Column I	Column II
(a) $2^3 \times 3^3$	(i) $\frac{1}{x^2}$
(b) $(xy)^2$	(ii) 6^3
(c) x^{-2}	(iii) x^2y^2
(d) $\left(\frac{2^4}{2^2}\right)^2$	(iv) 2^4

(A) a–iv, b–i, c–ii, d–iii
(B) a–ii, b–iii, c–i, d–iv
(C) a–i, b–ii, c–iii, d–iv
(D) a–ii, b–i, c–iv, d–iii